	School of Engineering & Technology	
	Department: SOET	**Session: 24-25 Even**
	Program: B. Tech CSE	**Semester: IV**
	Course Code: ENCS254	**Number of students:114**
	Course Name: Database Management Systems Lab	**Faculty: Dr. Reenu**

Lab Sheet 4

Lab Assignment: SQL User Management & Access Control

Title:

Implementation of **User Creation, Deletion, and Privilege Management in SQL**

Problem Statement

Design and execute SQL commands to **create users, manage their access privileges, and remove users** in a database system. This assignment focuses on **database security and access control mechanisms**.

```
CREATE DATABASE IF NOT EXISTS college_db;

USE college_db;
CREATE TABLE IF NOT EXISTS Student (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   age INT
);

CREATE TABLE IF NOT EXISTS Course (
   course_id INT PRIMARY KEY,
   course_name VARCHAR(50)
);
TRUNCATE TABLE Student;
INSERT INTO Student VALUES
(1,'Amit',20),
(2,'Neha',21);

TRUNCATE TABLE Course;
INSERT INTO Course VALUES
(101,'DBMS'),
(102,'OS');
```

Tasks

Part A: User Creation

1. Create a new user named `user1` with a password.

   : CREATE USER IF NOT EXISTS 'user1'@'localhost' IDENTIFIED BY 'pass123';

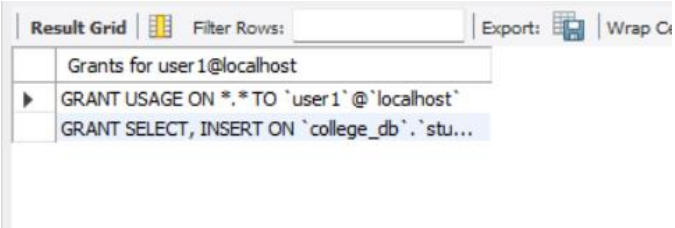
2. Create another user `user2` with limited access.

   : CREATE USER IF NOT EXISTS 'user2'@'localhost' IDENTIFIED BY 'pass456';

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table `Student` to `user1`.

   : GRANT SELECT, INSERT ON college_db.Student TO 'user1'@'localhost';



Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT, INSERT ON `college_db`.`stu...

4. Grant **ALL privileges** on a table `Course` to `user2`.

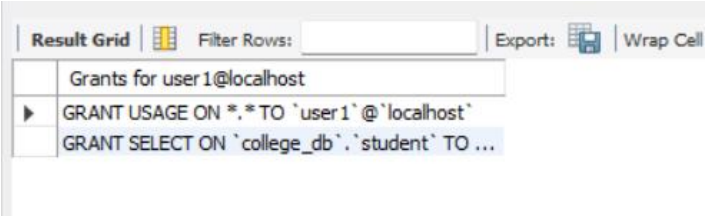
   : GRANT ALL PRIVILEGES ON college_db.Course TO 'user2'@'localhost';

5. Provide **WITH GRANT OPTION** to `user1` and observe its effect.

   : GRANT SELECT, INSERT ON college_db.Student

   TO 'user1'@'localhost'

   WITH GRANT OPTION;



Grants for user1@localhost
GRANT USAGE ON *.* TO `user1`@`localhost`
GRANT SELECT ON `college_db`.`student` TO ...

Part C: Revoking Privileges

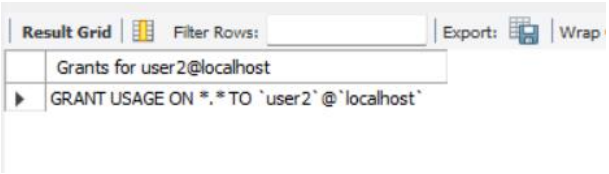
6. Revoke **INSERT permission** from `user1` on the `Student` table.

   : REVOKE INSERT ON college_db.Student FROM 'user1'@'localhost';

7. SHOW NTS FOR 'user1'@'localhost'; Revoke all privileges from `user2` on the `Course` table.

   : REVOKE ALL PRIVILEGES ON college_db.Course FROM 'user2'@'localhost';

   SHOW GRANTS FOR 'user2'@'localhost';



Grants for user2@localhost
GRANT USAGE ON *.* TO `user2`@`localhost`

Part D: User Deletion

8. Drop/Delete user `user2` from the database.

   DROP USER 'user2'@'localhost';

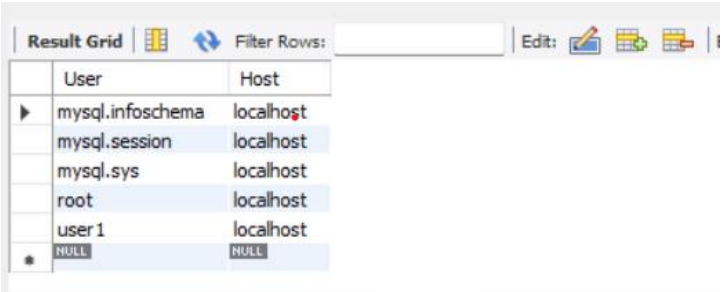
9. Attempt to login using deleted credentials and observe the result.

   Access denied for user 'user2'.

Part E: Additional Commands

10. Display all users in the system.

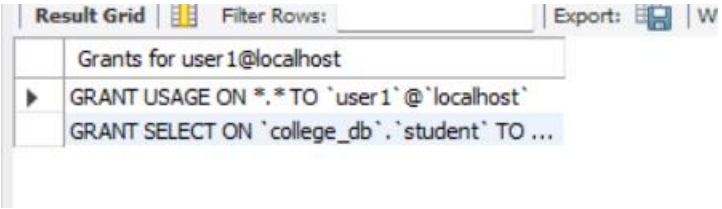
    SELECT User, Host FROM mysql.user;



User	Host
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost
user1	localhost
NULL	NULL

11. Show privileges assigned to a specific user.

SHOW GRANTS FOR 'user1'@'localhost';



Result Grid | Filter Rows: | Export: | W

	Grants for user1@localhost
▶	GRANT USAGE ON *.* TO `user1`@`localhost`
	GRANT SELECT ON `college_db`.`student` TO ...